

Industry trends

- Between 2026 and 2028 enterprises will increasingly adopt autonomous, AI-driven governance systems capable of automated policy enforcement, continuous data quality scoring, and real-time anomaly detection. Gartner forecasts that AI-driven automation will reduce manual data stewardship tasks by 40% by 2027.
- Governance models will shift from centralised to federated and hybrid, ultimately evolving towards autonomous domain-driven governance. Gartner reports that over 60% of enterprises will adopt federated governance by 2027.
- AI Trust, Risk, and Security Management (AI TRiSM) will become the top governance investment area as organisations confront risks related to hallucinations, bias, and regulatory compliance. Gartner predicts that enterprises implementing AI TRiSM will reduce AI-related risk incidents by 50% by 2026.
- According to Gartner, 75% of AI platforms will include built-in governance controls by 2027. Databricks Mosaic AI, Snowflake Cortex, and Microsoft Azure AI's Responsible AI Dashboard exemplify this trend.
- Synthetic data will become a regulated and essential component of AI training. Gartner projects that synthetic data will overshadow real data in AI training by 2030. McKinsey estimates that 50% of AI training datasets will include synthetic data by 2028.
- Global regulations will mandate transparency, lineage, and automated audits. Gartner states that regulatory pressure will be the top driver of data governance investments through 2030.
- Data contracts will replace traditional API documentation, enforcing schema, SLAs, lineage, and quality. Gartner predicts that data contracts will reduce integration failures by 40% by 2027.

prompt governance, LLM hallucinations and autonomous agent controls.

Governance alone does not create value. Enterprises need data capabilities that make governance operational while enabling innovation and AI adoption. Modern data ecosystems require intelligent platforms capable of discovering, understanding, protecting, and serving data on a scale.

Modern data architecture for AI

Modern data architecture provides the capabilities necessary for analytics, machine learning, generative AI, and autonomous systems. It enables enterprises to manage data as a strategic asset while ensuring governance, security, and scalability. It is a unified, governed, AI-ready data foundation that enables trusted insights, intelligent automation, and autonomous decision-making through reusable data products, continuous observability, and embedded governance controls.

The architecture is organised into two structural categories (Figure 1). The first five layers form the primary pipeline — the path data travels from the moment it is created in a source system to the moment it produces a business outcome.

The remaining three layers are cross-cutting disciplines which are applied continuously, at every stage, from ingestion through consumption. A modern AI-ready data architecture provides the infrastructure necessary for analytics, machine learning, generative AI, and autonomous systems. It enables organisations to manage data as a strategic asset while ensuring governance, security, and scalability.

Data sources: This layer represents the full surface area of enterprise data — every system, channel, partner relationship, and unstructured artifact that generates information the organisation can use. It groups the ecosystem into four major categories:

Data in silos: Data is segmented across various platforms, channels, tools, and business units, making it challenging to access it across the enterprise. It resides in ERP systems, CRM platforms, legacy applications, cloud-native platforms, data warehouses, data lakes, and SaaS applications. This leads to inefficiency, data duplication and data inconsistency.

Data accuracy: Completeness and timeliness of data is an issue.

Data quality: Lack of oversight of the quality of data coming into an enterprise as well as its usage throughout the organisation leads to poor data quality. Common quality challenges include missing values, duplicate records, outdated information,

inconsistent definitions, incomplete lineage and data drift.

Regulatory complexity: Managing regulatory compliance, and establishing data security and data privacy are challenges. Enterprises must comply with GDPR, HIPAA, CCPA, PCI-DSS, and the EU AI Act along with industry-specific regulations.

Data management: A poor data management strategy may lead to an enormous amount of data in a completely unmanageable format.

Data leakage: Leaks of sensitive business information or customer data can lead to data breaches.

AI-specific risks: New AI-era governance concerns include algorithmic bias, explainability requirements, training data provenance,